



Wrocław University
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Faculty of Pure and Applied Mathematics

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Master's Thesis

SELECTED STATISTICAL METHODS IN FMRI IMAGE DENOISING

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keywords:

tutaj podajemy najważniejsze słowa kluczowe w języku angielskim (łącznie nie powinny być dłuższe niż 150 znaków)

short summary:

The thesis focuses on comparison of selected statistical methods in functional Magnetic Resonance Imaging (fMRI) images denoising. The work focuses on proper data acquisition, correct implementation of methods and accurate comparison of denoised results using metrics commonly used in the field of fMRI. The goal is to find strengths and weaknesses of each method and evaluate the best for each case scenario.

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*For the purposes of archival thesis qualified to:**

a) category A (perpetual files)

b) category BE 50 (subject to expertise after 50 years)

** delete as appropriate*

stamp of the faculty

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Chapter 1

Introduction

Modern science combines a wide range of engineering and scientific methods to study processes occurring in the human brain. The development of Magnetic Resonance Imaging (MRI), together with methods for measuring changes related to brain activity, led to Functional Magnetic Resonance Imaging (fMRI).

MRI uses the magnetic properties of hydrogen nuclei, which are abundant in water and other molecules in the human body. A strong static magnetic field partially aligns the nuclear spins. Radio-frequency pulses then disturb this alignment, and after the pulse is switched off, the nuclei slowly return to their rest state. During this process, they generate signals that can be detected by receiver coils. Magnetic field gradients and image reconstruction algorithms are then used to determine the spatial origin of these signals and form an image of the body. *CITATION NEEDED: how MRI works physically*

However, both biological components and imaging are humanly imperfect. As a result, fMRI data contains several types of noise sources. These make it more difficult to estimate brain responses related to the studied experiments. Therefore researchers have developed a plethora of methods for reducing noise and improving the task-related Blood-Oxygen-Level-Dependent (BOLD) signals.

The BOLD signal is related to changes in blood oxygenation caused by the brain's response to neural activity, called haemodynamic response. When neural activity increases in a given area, it is then followed by changes in local blood flow, blood volume, and oxygen consumption. These processes directly influence the measured MR signal. The BOLD response is not a direct or one-to-one measurement of neural activity, rather it is an indirect and delayed physiological response, but its changes can still be used to study how the brain responds to experimental conditions [1]. *CITATION NEEDED: BOLD contrast, haemodynamic response (also function?)*

The main sources of unwanted variability include:

- scanner- and acquisition-related noise, such as thermal noise, electronic noise, low-frequency signal drift,
- movement-related effects, especially head motion,
- physiological noise caused by natural processes in the human body, including respiration, cardiac activity, and changes in blood flow,
- preprocessing and image-processing artifacts, which are introduced during motion correction, registration, etc.

CITATION NEEDED: classification of sources of noise in fMRI

Several families of methods have been developed for reducing noise in fMRI data:

- regression-based methods,
- component-based methods,
- frequency-domain methods,
- wavelet-based methods,
- spatial and temporal regularization methods,
- partial differential equation-based methods,
- machine-learning methods.

CITATION NEEDED: examples of methods for each family

In practice, fMRI preprocessing and denoising pipelines often combine several methods because different procedures address different sources of noise. The appropriate combination of methods needs to be well documented and explained, since overly aggressive denoising may remove part of the task-related signal. The effectiveness of a denoising method must therefore be considered together with the risk of signal loss and model overfitting.

CITATION NEEDED: combined denoising methods + signal loss caused by too aggressive denoising

This thesis focuses solely on statistical methods. The mathematical models, that will be utilized in this thesis are:

- General Linear Model (GLM),
- Principal Component Analysis (PCA),
- Independent Component Analysis (ICA),
- Least Absolute Shrinkage and Selection Operator (LASSO).

Denoising procedure will be using above-mentioned methods in different combination in order to compare approaches, methodologies and analyze them in the context of one-back repetition-detection task fMRI dataset, further described in chapter 2. These approaches are further described in chapter 3 and are as follows:

- Standard nuisance GLM,
- PCA-assisted GLM,
- ICA-based denoising,
- LASSO nuisance regression.

The main reason for considering these approaches is that they can be described using related linear-algebraic frameworks. Although crucially, they differ in their assumptions about the structure of the data, the construction of the model and the properties of underlying components. For example, different methods require different assumptions related to orthogonality or statistical independence. Nevertheless this narrower selection

will make the comparison easier to interpret and methodologically consistent. *CITATIONS NEEDED: assumptions of the individual methods*

This choice was made because methods based on wavelet transforms, partial differential equations or other methods, require different mathematical tools and may also require different evaluation criteria. Including them would considerably broaden the scope of the thesis and could weaken the main line of comparison.

The aim of the thesis is not to determine which statistical method is universally the best - such conclusion would strongly depend on the dataset, experiments being carried out, noise structure, parameter selection, and evaluation criteria. Instead, the thesis investigates how selected regression- and component-based methods influence denoising quality. Particular attention is given to how the assumptions made by GLM-based approaches, PCA, ICA, and LASSO affect signal preservation and noise removal.

Chapter 2

Dataset overview

2.1 Dataset description and experimental framework

2.1.1 Dataset description

The dataset used in this thesis is the *Visual Object Recognition fMRI dataset*, available through OpenNeuro under accession number `ds000105` (formerly `ds105`). The version considered here is snapshot 3.0.0 [2]. The dataset contains functional and high-resolution anatomical images, together with event files describing the presented stimuli, their durations, and their onset times. The anatomical images were defaced to reduce the risk of participant re-identification.

The data was used in three related studies concerning the representation of faces and objects in the ventral temporal cortex of the human brain. The first study investigated whether viewing different object categories produced distinct patterns of fMRI activity [3]. The second study examined this distributed representation hypothesis using a neural network [4]. The third study investigated whether similarities between neural response patterns reflected visual similarities between the presented images [5].

Taken together, these studies aimed to characterize object representations in the ventral temporal cortex as distributed and overlapping rather than restricted to separate regions for individual object categories.

The dataset contains measurements from six participants, including five females and one male. Twelve functional runs were acquired for each participant.

2.1.2 Experimental task and design

During fMRI data acquisition, the participants performed a one-back repetition-detection task while viewing grayscale images belonging to the following visual categories:

1. faces,
2. houses,
3. cats,
4. bottles,
5. scissors,
6. shoes,

7. chairs,
8. nonsense images.

Examples of the presented stimuli are shown in Figure 2.1, which is taken from the original paper [3].

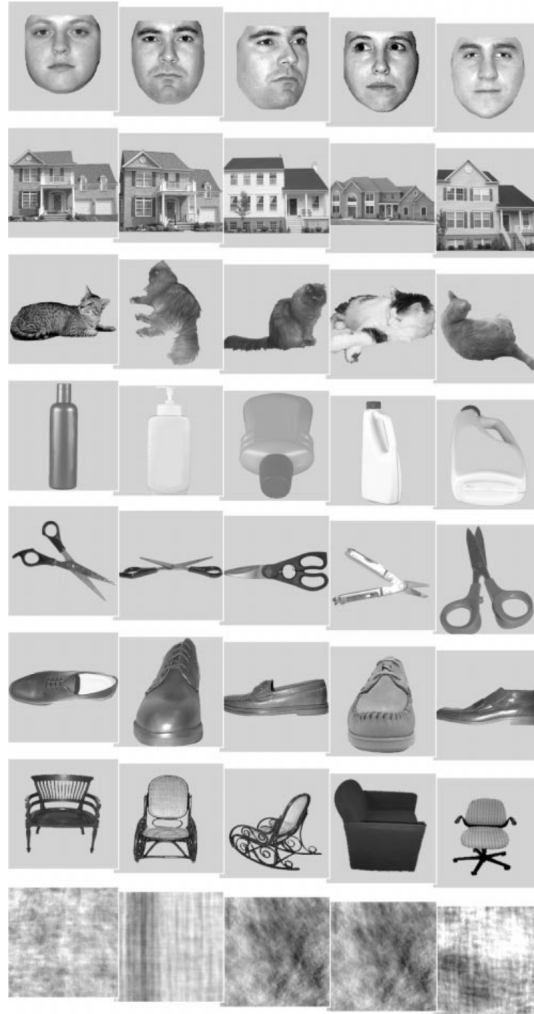


Figure 2.1: Examples of the presented visual categories.

The experiment followed a block design. Each run contained eight stimulus blocks, with one block corresponding to each visual category. Each block contained 12 image presentations. An image was displayed for 500 ms and was followed by a rest interval of 1500 ms. Therefore, each presentation lasted 2 seconds, and each category block - 24 seconds.

The stimulus blocks were separated by 12-second rest periods. Each run also began and ended with 12 seconds of rest. Consequently, the complete experimental sequence lasted 300 seconds.

A total of 96 images were presented during each run. The same eight categories were included in all 12 runs, although the order of the category blocks differed between runs. The block structure itself remained unchanged.

2.1.3 Functional MRI acquisition

The functional data were acquired using gradient-echo echo-planar imaging on a GE 3T scanner. The main acquisition parameters are presented in Table 2.1 [2].

Parameter	Value
Repetition time, TR	2500 ms
Echo time, TE	30 ms
Flip angle	90°
Number of slices	40
Slice thickness	3.5 mm
Field of view	24 cm
Number of time points per run	121
Number of functional runs per participant	12

Table 2.1: Functional MRI acquisition parameters.

The functional images have spatial dimensions of $40 \times 64 \times 64$. Each run contains 121 functional volumes and covers approximately five minutes of acquisition.

The repetition time of 2500 ms means that the BOLD signal is not sampled separately for every individual image presentation, since the images were presented every 2 seconds. The stimulus timing will be represented by task regressors used in design matrix for General Linear Model (further explained in later section LATER ADD REFERENCE), convolved with a hemodynamic response function (HRF) to model the expected BOLD response [6].

2.1.4 Anatomical MRI acquisition

The dataset also contains one high-resolution T1-weighted anatomical image for each participant. These images consist of 124 slices with a slice thickness of 1.2 mm and a field of view of 24 cm. Their spatial dimensions are $124 \times 256 \times 256$.

The anatomical images provide greater spatial detail than the functional images. In this thesis, they are used as an anatomical reference, for brain extraction, for registration of the functional scans, and for visualization of statistical maps. They are not subjected to the denoising methods considered in the thesis.

2.2 Organization of the dataset

The dataset is organized according to the Brain Imaging Data Structure (BIDS) convention [7]. At the highest level, it contains metadata describing the experiment and separate directories corresponding to individual participants. Each participant directory contains an **anat** directory (T1-weighted anatomical data) and a **func** directory (functional data).

For each participant, the **func** directory contains 12 functional BOLD images and 12 corresponding event files. Each BOLD file is a four-dimensional image with shape:

$$X \times Y \times Z \times T, \quad (2.1)$$

where X , Y , and Z represent the spatial dimensions and T represents the number of time points.

The event files contain the columns `onset`, `duration`, and `trial_type`. The `onset` value gives the time of a stimulus presentation in seconds relative to the beginning of the corresponding run. The `duration` value specifies how long the stimulus was displayed, while `trial_type` identifies its category.

2.3 Caveats related to the dataset

The experimental conditions were repeated across 12 runs. This makes leave-one-run-out cross-validation possible. A model can be fitted using 11 runs and evaluated on the remaining run, with the procedure repeated for each run. This allows for the training-test split and clearer evaluation of the of the fitted model and can support the selection of model parameters. The multiple-run structure may also help identify run-specific noise effects.

The dataset does not provide a noise-free reference image or a known ground-truth BOLD signal. The quality of denoising must therefore be evaluated indirectly using criteria such as:

- prediction accuracy on left-out runs,
- changes in residual variance, preferably evaluated on left-out data,
- changes in temporal signal-to-noise ratio (tSNR) measures,
- preservation and stability of expected task-related responses,
- reduction of anatomically impossible spatial patterns.

The dataset has also a few limitations. It contains only six participants and was acquired using an older method with relatively coarse spatial and temporal resolution. It represents one specific visual experiment with a block design. Therefore conclusions obtained using this dataset may not generalize directly to other conditions (like event-related designs, resting-state data, or experiments involving different processes). Furthermore, the dataset does not include separate respiratory or cardiac recordings that could be used to construct physiological nuisance regressors.

Nevertheless, the repeated and clearly defined experimental structure makes the dataset useful for comparing denoising methods. In particular, the presence of 12 runs per participant supports within-subject cross-validation and the evaluation of the stability of task-related estimates.

Chapter 3

Statistical foundations of the selected denoising methods

Several statistical methods are considered in this thesis:

1. General Linear Model (GLM),
2. Independent Component Analysis (ICA),
3. Principal Component Analysis (PCA),
4. lasso,
5. basis pursuit.

The final comparison may include only a subset of these methods. In particular, the inclusion of the lasso, basis pursuit, or PCA will depend on their suitability for the denoising framework developed in the later chapters.

An important motivation for selecting these methods is that fMRI noise may have temporal and spatial structure. Noise patterns may be correlated across time points, voxels, or both. Therefore, the data should not always be treated as a collection of completely independent measurements. The selected approaches use relationships between observations, regressors, or latent components to estimate task-related signal and unwanted variability.

This thesis focuses on methods that can be described using statistical models and matrix operations. The main interest is the estimation or separation of signal and noise components. Although the methods differ in their assumptions and objectives, they can be discussed using a related mathematical language.

A common starting point is the linear model

$$Y = XB + E, \tag{3.1}$$

where Y denotes the observed data, X is the design matrix, B contains the unknown coefficients, and E represents the part of the data not explained by the model.

A more general matrix factorization can be written as

$$Y \approx DA, \tag{3.2}$$

where D and A represent two matrices whose interpretation depends on the method. Different approaches impose different assumptions or restrictions on these matrices, their

coefficients, latent components, or error terms. These may include assumptions related to orthogonality, statistical independence, sparsity, or the structure of residual noise.

The possibility of describing the selected methods within a related mathematical framework is one of the main reasons for their selection. Their comparison allows the following aspects to be studied:

- the role of the design matrix;
- the role of nuisance regressors;
- dimensionality reduction;
- separation of signal and noise;
- regularization;
- stability of parameter estimates;
- selection and optimization of hyperparameters.

The following sections introduce the mathematical foundations of each method, the assumptions on which it is based, and its possible role in fMRI denoising.

3.1 General Linear Model

3.1.1 Model definition

The first model is the General Linear Model (GLM). It is viewed as a special case of the Generalized Linear Model with a Gaussian response distribution and the identity link function, defined in [8, Chapter 1.1.4] as Linear Model. However, the name General Linear Model is conventionally used in the fMRI literature.

In general for a single voxel in fMRI, the GLM can be written as

$$\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}, \quad (3.3)$$

where

- $\mathbf{y} \in \mathbb{R}^n$ contains the observed BOLD signal at the voxel over n time points;
- $\mathbf{X} \in \mathbb{R}^{n \times p}$ is the design matrix;
- $\boldsymbol{\beta} \in \mathbb{R}^p$ contains the unknown regression coefficients;
- $\boldsymbol{\varepsilon} \in \mathbb{R}^n$ is the error vector.

Each column of $\mathbf{X}$ represents a task-related effect or another modeled source of variability. The corresponding element of $\boldsymbol{\beta}$ describes the estimated contribution of that regressor to the BOLD time series at the considered voxel. In a voxel-wise fMRI analysis, the same design matrix is usually applied separately to each voxel, while a different coefficient vector is estimated for each voxel.

Equivalently, the models for all voxels can be written simultaneously as $\mathbf{Y} = \mathbf{X}\mathbf{B} + \mathbf{E}$, where each column of $\mathbf{Y}$ contains the time series of one voxel.

In the Linear Model described in [8, Chapter 2], the error term is commonly assumed to satisfy

$$\mathbb{E}(\boldsymbol{\varepsilon}) = \mathbf{0}, \quad \text{Var}(\boldsymbol{\varepsilon}) = \sigma^2 \mathbf{I}. \quad (3.4)$$

The normality of the error term is not required under the Gauss-Markov assumptions [8, Gauss-Markov theorem].

3.1.2 Ordinary Least Squares estimation

The Ordinary Least Squares (OLS) estimator is obtained by minimizing the objective function L , which is equal to squared Euclidean norm of the residual vector:

$$L(\boldsymbol{\beta}) = \|\mathbf{y} - \mathbf{X}\boldsymbol{\beta}\|_2^2 \quad (3.5)$$

$$= (\mathbf{y} - \mathbf{X}\boldsymbol{\beta})^T (\mathbf{y} - \mathbf{X}\boldsymbol{\beta}) \quad (3.6)$$

$$= \mathbf{y}^T \mathbf{y} - 2\mathbf{y}^T \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\beta}^T \mathbf{X}^T \mathbf{X}\boldsymbol{\beta}. \quad (3.7)$$

To minimize this function, its derivative with respect to $\boldsymbol{\beta}$ is set equal to zero:

$$\frac{\partial L(\boldsymbol{\beta})}{\partial \boldsymbol{\beta}} = -2\mathbf{X}^T (\mathbf{y} - \mathbf{X}\boldsymbol{\beta}). \quad (3.8)$$

For the estimator $\hat{\boldsymbol{\beta}}$, this gives the normal equations [8, eq. 2.2]

$$\mathbf{X}^T \mathbf{y} = \mathbf{X}^T \mathbf{X} \hat{\boldsymbol{\beta}}. \quad (3.9)$$

If $\mathbf{X}$ has full column rank p , then $\mathbf{X}^T \mathbf{X}$ is invertible and the OLS estimator is

$$\hat{\boldsymbol{\beta}}_{\text{OLS}} = (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \mathbf{y}. \quad (3.10)$$

The fitted values are then given by

$$\hat{\mathbf{y}} = \mathbf{X} \hat{\boldsymbol{\beta}}_{\text{OLS}} \quad (3.11)$$

$$= \mathbf{X} (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \mathbf{y}. \quad (3.12)$$

We can construct the Hat matrix [9], that describes the transformation $\hat{\mathbf{y}} = \mathbf{H}\mathbf{y}$:

$$\mathbf{H} = \mathbf{X} (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T. \quad (3.13)$$

The residual vector is therefore:

$$\mathbf{e} = \mathbf{y} - \hat{\mathbf{y}} \quad (3.14)$$

$$= \mathbf{y} - \mathbf{H}\mathbf{y} \quad (3.15)$$

$$= (\mathbf{I} - \mathbf{H}) \mathbf{y}. \quad (3.16)$$

Because

$$\begin{aligned}
(\mathbf{I} - \mathbf{H}) \mathbf{X} &= \left(\mathbf{I} - \mathbf{X} (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \right) \mathbf{X} \\
&= \left(\mathbf{X} - \mathbf{X} (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \mathbf{X} \right) \\
&= \left(\mathbf{X} - \mathbf{X} (\mathbf{X}^T \mathbf{X})^{-1} (\mathbf{X}^T \mathbf{X}) \right) \\
&= (\mathbf{X} - \mathbf{X} \mathbf{I}) = \mathbf{0},
\end{aligned} \tag{3.17}$$

the residual vector in Equation 3.16 can also be written as

$$\mathbf{e} = (\mathbf{I} - \mathbf{H})(\mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}) = (\mathbf{I} - \mathbf{H})\boldsymbol{\varepsilon}. \tag{3.18}$$

The fitted values and residuals do not by themselves define the denoised fMRI signal. For denoising, the columns of the design matrix must first be divided into task-related and nuisance regressors. The estimated nuisance contribution can then be subtracted from the observed signal. This procedure, together with the construction of the design matrix, will be described in the later sections.

3.1.3 Generalized Least Squares and prewhitening

The standard OLS model assumes that the observations have equal error variances and that their errors are uncorrelated:

$$\text{Var}(\boldsymbol{\varepsilon}) = \sigma^2 \mathbf{I} \quad \wedge \quad \forall_{i \neq j} \quad \text{Var}(\varepsilon_i, \varepsilon_j) = 0. \tag{3.19}$$

This assumption is generally not possible for fMRI time series. Measurements from the same voxel at consecutive time points are commonly autocorrelated. Therefore, the covariance matrix of the errors may have a more general form [8, Chapter 2.7.2]:

$$\text{Var}(\boldsymbol{\varepsilon}) = \sigma^2 \mathbf{V}, \tag{3.20}$$

where $\mathbf{V}$ describes the temporal correlation structure and doesn't have to be the identity matrix.

Suppose that

- $\mathbf{X}$ has full column rank;
- $\mathbf{V}$ is a known, positive definite matrix.

The spectral decomposition of $\mathbf{V}$ is

$$\mathbf{V} = \mathbf{Q} \boldsymbol{\Lambda} \mathbf{Q}^T, \tag{3.21}$$

where $\boldsymbol{\Lambda}$ is a diagonal matrix containing the positive eigenvalues of $\mathbf{V}$ and $\mathbf{Q}$ is an orthogonal matrix whose columns are the corresponding eigenvectors:

$$\mathbf{Q}^T \mathbf{Q} = \mathbf{Q} \mathbf{Q}^T = \mathbf{I}. \tag{3.22}$$

The symmetric square root and inverse square root of $\mathbf{V}$ can therefore be defined as

$$\mathbf{V}^{1/2} = \mathbf{Q}\mathbf{\Lambda}^{1/2}\mathbf{Q}^T, \quad (3.23)$$

$$\mathbf{V}^{-1/2} = \mathbf{Q}\mathbf{\Lambda}^{-1/2}\mathbf{Q}^T. \quad (3.24)$$

The transformed variables are defined as

$$\mathbf{y}^* = \mathbf{V}^{-1/2}\mathbf{y}, \quad \mathbf{X}^* = \mathbf{V}^{-1/2}\mathbf{X}, \quad \boldsymbol{\varepsilon}^* = \mathbf{V}^{-1/2}\boldsymbol{\varepsilon}. \quad (3.25)$$

The transformed model is

$$\mathbf{y}^* = \mathbf{X}^*\boldsymbol{\beta} + \boldsymbol{\varepsilon}^*. \quad (3.26)$$

Its expected value is

$$\mathbb{E}(\mathbf{y}^*) = \mathbb{E}(\mathbf{V}^{-1/2}\mathbf{y}) \quad (3.27)$$

$$= \mathbf{V}^{-1/2}\mathbf{X}\boldsymbol{\beta} \quad (3.28)$$

$$= \mathbf{X}^*\boldsymbol{\beta}. \quad (3.29)$$

The covariance matrix of the transformed errors is

$$\text{Var}(\boldsymbol{\varepsilon}^*) = \text{Var}(\mathbf{V}^{-1/2}\boldsymbol{\varepsilon}) \quad (3.30)$$

$$= \mathbf{V}^{-1/2} \text{Var}(\boldsymbol{\varepsilon}) (\mathbf{V}^{-1/2})^T \quad (3.31)$$

$$= \mathbf{V}^{-1/2} \sigma^2 \mathbf{V} (\mathbf{V}^{-1/2})^T \quad (3.32)$$

$$= \sigma^2 \mathbf{V}^{-1/2} \mathbf{V} \mathbf{V}^{-1/2} \quad (3.33)$$

$$= \sigma^2 \mathbf{V}^{-1/2} \mathbf{V}^{1/2} \mathbf{V}^{1/2} \mathbf{V}^{-1/2} \quad (3.34)$$

$$= \sigma^2 \mathbf{I}. \quad (3.35)$$

Thus, the transformed errors are uncorrelated and have equal variance. OLS can therefore be applied to the transformed model. This gives

$$\hat{\boldsymbol{\beta}}_{\text{GLS}} = [(\mathbf{X}^*)^T \mathbf{X}^*]^{-1} (\mathbf{X}^*)^T \mathbf{y}^* \quad (3.36)$$

$$= \left[(\mathbf{V}^{-1/2} \mathbf{X})^T \mathbf{V}^{-1/2} \mathbf{X} \right]^{-1} (\mathbf{V}^{-1/2} \mathbf{X})^T \mathbf{V}^{-1/2} \mathbf{y} \quad (3.37)$$

$$= [\mathbf{X}^T \mathbf{V}^{-1/2} \mathbf{V}^{-1/2} \mathbf{X}]^{-1} \mathbf{X}^T \mathbf{V}^{-1/2} \mathbf{V}^{-1/2} \mathbf{y} \quad (3.38)$$

$$= (\mathbf{X}^T \mathbf{V}^{-1} \mathbf{X})^{-1} \mathbf{X}^T \mathbf{V}^{-1} \mathbf{y}. \quad (3.39)$$

This is the Generalized Least Squares (GLS) estimator of $\boldsymbol{\beta}$ [8, Equation 2.12].

The transformation of the data and design matrix by $\mathbf{V}^{-1/2}$ is referred to as *prewhitening* in neuroimaging. If the covariance structure $\mathbf{V}$ is correctly specified and known, GLS provides the Best Linear Unbiased Estimator (BLUE) of the regression coefficients [8][10][11].

In practice, $\mathbf{V}$ is unknown and must be estimated from the data. From this, the resulting procedure is described as Feasible Generalized Least Squares (FGLS). Estimation of the temporal covariance structure, including its possible autoregressive representation, will be discussed in a later section.

Covariance of the prewhitened residuals

Let the hat matrix derived from Equation 3.13 for the transformed model be

$$\mathbf{H}^* = \mathbf{X}^* \left[(\mathbf{X}^*)^T \mathbf{X}^* \right]^{-1} (\mathbf{X}^*)^T. \quad (3.40)$$

The residual vector for the transformed model is

$$\mathbf{e}^* = (\mathbf{I} - \mathbf{H}^*) \mathbf{y}^* \quad (3.41)$$

$$= (\mathbf{I} - \mathbf{H}^*) (\mathbf{X}^* \boldsymbol{\beta} + \boldsymbol{\varepsilon}^*) \quad (3.42)$$

$$= (\mathbf{I} - \mathbf{H}^*) \mathbf{X}^* \boldsymbol{\beta} + (\mathbf{I} - \mathbf{H}^*) \boldsymbol{\varepsilon}^*. \quad (3.43)$$

The first term is equal to zero from Equation 3.17. Therefore,

$$\mathbf{e}^* = (\mathbf{I} - \mathbf{H}^*) \boldsymbol{\varepsilon}^*. \quad (3.44)$$

The covariance matrix of the transformed residuals is then

$$\text{Var}(\mathbf{e}^*) = (\mathbf{I} - \mathbf{H}^*) \text{Var}(\boldsymbol{\varepsilon}^*) (\mathbf{I} - \mathbf{H}^*)^T \quad (3.45)$$

$$= \sigma^2 (\mathbf{I} - \mathbf{H}^*) (\mathbf{I} - \mathbf{H}^*)^T. \quad (3.46)$$

The hat matrix $\mathbf{H}^*$ is symmetric and idempotent [9]:

$$(\mathbf{H}^*)^T = \mathbf{H}^*, \quad (\mathbf{H}^*)^2 = \mathbf{H}^*. \quad (3.47)$$

Therefore, $\mathbf{I} - \mathbf{H}^*$ is also symmetric and idempotent, and

$$\text{Var}(\mathbf{e}^*) = \sigma^2 (\mathbf{I} - \mathbf{H}^*). \quad (3.48)$$

This is the theoretical covariance matrix of the residuals obtained after prewhitening and model fitting. The transformed errors have covariance $\sigma^2 \mathbf{I}$, whereas the transformed residuals have covariance $\sigma^2 (\mathbf{I} - \mathbf{H}^*)$ because they have additionally been projected onto the residual subspace.

Other articles propose additionally high-pass filter to address low-frequency signal drift [10]. In this thesis, low-frequency trends will instead be represented by drift regressors included in the design matrix. The construction of these regressors and their relationship to temporal filtering will be discussed in the later sections.

3.2 Autoregressive temporal-noise model

As mentioned in Section 3.1.3, the assumption of uncorrelated errors is generally not appropriate for fMRI time series. This creates the need to for the covariance matrix $\mathbf{V}$ introduced in Equation 3.20. Since $\mathbf{V}$ is unknown, it must be estimated from the observed data [8, Chapter 2.7.3].

In time-series data, the regression errors are unlikely to be independent, especially in fMRI data. In this thesis they are assumed to be a stationary process, which satisfies:

$$\mathbb{E}[\varepsilon_t] = 0, \quad \text{Var}[\varepsilon_t] = \gamma(0) = \sigma_\varepsilon^2, \quad (3.49)$$

and

$$\text{Cov}[\varepsilon_t, \varepsilon_{t+s}] = \gamma(s), \quad \rho(s) = \frac{\gamma(s)}{\gamma(0)}, \quad (3.50)$$

where $\gamma(s)$ is the autocovariance function and $\rho(s)$ is the autocorrelation function at lag s .

The covariance matrix of the error vector can then be written as

$$\mathbf{\Sigma} = \sigma_\varepsilon^2 \mathbf{V}, \quad V_{ij} = \rho(|i - j|). \quad (3.51)$$

Thus, $\mathbf{V}$ is the temporal correlation matrix, while $\mathbf{\Sigma}$ is the corresponding covariance matrix.

A simple model for temporally autocorrelated regression errors is the first-order autoregressive process, denoted by AR(1) [12]:

$$\varepsilon_t = \phi \varepsilon_{t-1} + \nu_t, \quad \nu_t \stackrel{\text{i.i.d.}}{\sim} \mathcal{N}(0, \sigma_\nu^2), \quad (3.52)$$

where ϕ is the autoregressive coefficient and ν_t is the noise term.

For the AR(1) process to be stationary, the autoregressive coefficient must satisfy

$$|\phi| < 1. \quad (3.53)$$

Under this condition, the variance and autocorrelation function are

$$\sigma_\varepsilon^2 = \frac{\sigma_\nu^2}{1 - \phi^2}, \quad \rho(s) = \phi^{|s|}. \quad (3.54)$$

In particular,

$$\rho(1) = \phi, \quad \rho(s) \rightarrow 0 \quad \text{as } |s| \rightarrow \infty. \quad (3.55)$$

The temporal correlation matrix therefore has the form

$$V_{ij} = \phi^{|i-j|}. \quad (3.56)$$

Since the true regression errors ε are not observed, the autoregressive coefficient must be estimated from residuals obtained from an initial OLS model fit, introduced in Equation 3.16:

$$\mathbf{e} = \mathbf{y} - \hat{\mathbf{y}} = \mathbf{y} - \mathbf{X}\hat{\boldsymbol{\beta}}_{\text{OLS}} \quad (3.57)$$

In order to estimate the autoregressive parameter ϕ we can use OLS to the AR(1) equation, treating e_t as the response and e_{t-1} as the predictor. The next step is therefore to minimize:

$$L(\phi) = \sum_{t=2}^n (e_t - \phi e_{t-1})^2, \quad (3.58)$$

finally obtaining:

$$\hat{\phi} = \frac{\sum_{t=2}^n e_t e_{t-1}}{\sum_{t=2}^n e_{t-1}^2}. \quad (3.59)$$

The estimated temporal correlation matrix is then constructed from the estimated autoregressive parameter

$$\hat{V}_{ij} = \hat{\phi}^{|i-j|}. \quad (3.60)$$

This estimated covariance matrix $\hat{\mathbf{V}}$ can now be used in the prewhitening transformation from Equation 3.25:

$$\mathbf{y}^* = \hat{\mathbf{V}}^{-1/2} \mathbf{y}, \quad \mathbf{X}^* = \hat{\mathbf{V}}^{-1/2} \mathbf{X}. \quad (3.61)$$

The regression model is then fitted again using the transformed data, which was thoroughly described in Section 3.1.3.

3.3 Construction of the design matrix

The construction of the design matrix is a crucial part of GLM-based denoising. The design matrix contains explanatory variables representing task-related and nuisance variability. The estimated coefficients of the task-related regressors are used to model the BOLD response associated with the experiment, while the coefficients of the nuisance regressors describe contribution of modeled nuisance.

For a single voxel, the GLM can be written as

$$\mathbf{y} = \mathbf{X}_{\text{task}} \boldsymbol{\beta}_{\text{task}} + \mathbf{X}_{\text{nuis}} \boldsymbol{\beta}_{\text{nuis}} + \boldsymbol{\varepsilon}, \quad (3.62)$$

where the complete design matrix and coefficient vector are partitioned as

$$\mathbf{X} = \begin{bmatrix} \mathbf{X}_{\text{task}} & \mathbf{X}_{\text{nuis}} \end{bmatrix}, \quad \boldsymbol{\beta} = \begin{bmatrix} \boldsymbol{\beta}_{\text{task}} \\ \boldsymbol{\beta}_{\text{nuis}} \end{bmatrix}. \quad (3.63)$$

After fitting the model, the estimated nuisance contribution is

$$\hat{\mathbf{y}}_{\text{nuis}} = \mathbf{X}_{\text{nuis}} \hat{\boldsymbol{\beta}}_{\text{nuis}}. \quad (3.64)$$

The denoised time series can then be defined by subtracting this contribution from the observed signal:

$$\mathbf{y}_{\text{denoised}} = \mathbf{y} - \hat{\mathbf{y}}_{\text{nuis}}. \quad (3.65)$$

This procedure removes only the contribution assigned to nuisance regressors, while the fitted modeled task-related component is not subtracted from the data.

A typical fMRI design matrix contains:

- task-related regressors;
- motion regressors;
- physiological regressors;
- drift regressors;
- data-derived nuisance regressors.

For a given run, the untransformed design matrix is the same for all analyzed voxels. Each voxel has its own estimated coefficient vector. Separate design matrices are constructed for different runs because stimulus timing and run-specific nuisance effects may differ between them.

3.3.1 Task-related regressors

As described in Chapter 2, the participants were presented with visual stimuli at specified times during each experimental run. The purpose of the task-related regressors is to model the expected BOLD response associated with these stimulus presentations.

The event files provided with the dataset contain the onset, duration, and category of each stimulus. The stimuli belong to eight visual categories. Therefore, the task-related part of the design matrix contains eight columns, with one column corresponding to each category:

$$\mathbf{X}_{\text{task}} \in \mathbb{R}^{n \times 8}, \quad (3.66)$$

where n is the number of data points in each run.

For category k , let J_k denote the number of stimulus presentations, and let o_{kj} and d_{kj} denote the onset and duration of j -th presentation. The stimulus function can then be written as

$$s_k(t) = \sum_{j=1}^{J_k} \mathbb{1}_{[o_{kj}, o_{kj}+d_{kj})}(t). \quad (3.67)$$

In the analyzed dataset, each category is presented 12 times within its block, and each individual image is displayed for 0.5 seconds. Thus, each category function is a sum of twelve 0.5-second boxcar functions separated by the corresponding interstimulus intervals.

The set of categories is the same for every run, but the order of the category blocks differs. As a result, the eight columns have the same interpretation across runs but unique values.

Hemodynamic response function

The stimulus functions describe when the stimuli were presented, but they do not directly represent the expected BOLD response. The actual haemodynamic response is delayed and temporally dispersed relative to the neural event currently taking place. Therefore, each stimulus function is convolved with a haemodynamic response function (HRF) [13].

The task regressor for category k [13, Equation 1] is defined as

$$x_k(t) = (s_k * \tilde{h})(t) + \eta(t) \quad (3.68)$$

where

1. $\tilde{h}$ denotes the selected HRF,
2. η denotes the noise of the measurement.

Since the objective is to model the clean BOLD signal in this regressor, the noise measurement will not be present, i.e. $\eta \equiv 0$, and instead will be present in the nuisance regressors and residuals.

The HRF describes the expected haemodynamic response following a short neural event. Its shape can be characterized by quantities such as its peak amplitude, time-to-peak, and full width at half maximum. Correctly modeled HRF should have its parameters reflect these key components of the haemodynamic response [14]:

- *amplitude*: magnitude of the BOLD response,

- *time-to-peak*: latency of the effect of stimulus,
- *full-width at half-maximum*: duration of neural activity.

Importantly, the relationship between neural activity and the measured BOLD response is not exact. Neural responses vary across presentations, while the HRF differs between participants, brain regions, and experimental conditions. The BOLD signal is also indirect and delayed reponse to neural activity. A fixed HRF may therefore differ from the true response present in the data [14].

Several approaches can be used to model the HRF. Additional basis functions or estimated HRF parameters provide greater flexibility, but they also increase the number of parameters, reduce the available residual degrees of freedom, and may increase the variance of the estimated coefficients [14]. To keep the model consistent across runs and methods, this thesis uses a single canonical HRF.

The selected function is the canonical HRF used in Statistical Parameter Mapping (SPM). A representation of this function is a difference of two gamma density functions [14, Equation A1]:

$$h(t) = g(t; \alpha_1, \theta_1) - cg(t; \alpha_2, \theta_2), \quad (3.69)$$

where

$$g(t; \alpha, \theta) = \frac{t^{\alpha-1} e^{-t/\theta}}{\Gamma(\alpha) \theta^\alpha}. \quad (3.70)$$

For the conventional double-gamma specification considered here,

$$\alpha_1 = 6, \quad \alpha_2 = 16, \quad \theta_1 = \theta_2 = 1, \quad c = \frac{1}{6}. \quad (3.71)$$

The first component represents the main positive haemodynamic response. The smaller and more delayed component represents the post-stimulus undershoot. These parameters are fixed rather than estimated separately from the analyzed data. This provides a robust and simple implementation, without the increased risk of inappropriate estimation.

For a fixed scaling convention, the HRF is normalized to have a maximum value of one:

$$\tilde{h}(t) = \frac{h(t)}{\max_{\tau \geq 0} h(\tau)}. \quad (3.72)$$

This normalization fixes the scale of the HRF and affects the interpretation of the corresponding regression coefficients. It does not change the temporal shape of the regressor.

The same HRF convolution is performed for each visual category. An example showing the stimulus function, canonical HRF, and resulting convolved task regressor is presented in Figure 3.1.

3.3.2 Nuisance regressors

The remaining part of the design matrix contains nuisance regressors. These regressors model structured sources of variability that are not considered part of the task-related response. They do not reproduce all forms of noise exactly. Instead, they provide explanatory variables whose fitted contributions can be removed from the observed signal.

The nuisance part of the design matrix may be written as

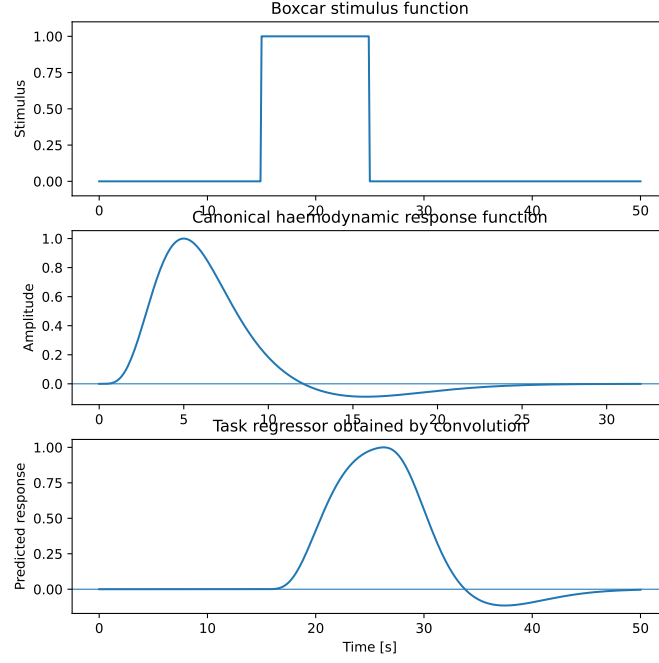


Figure 3.1: Example showing the stimulus function active in time period $[15.0s25.0s]$, canonical HRF and the convolution of both functions.

$$\mathbf{X}_{\text{nuis}} = \begin{bmatrix} \mathbf{X}_{\text{intercept}} & \mathbf{X}_{\text{drift}} & \mathbf{X}_{\text{motion}} & \mathbf{X}_{\text{physio}} & \mathbf{X}_{\text{data}} \end{bmatrix}. \quad (3.73)$$

Not all of these matrices must be present in every analysis. Their inclusion depends on the available measurements and the selected denoising procedure. Data-derived nuisance regressors based on PCA will be introduced in a later section.

Motion and physiological regressors

Motion regressors are commonly included in first-level fMRI models to account for signal changes associated with participant movement. A standard set contains three translation parameters and three rotation parameters estimated during motion correction.

The raw dataset does not provide precomputed motion parameters, hence explicit motion regressors cannot be included in the design matrix.

Physiological regressors may be constructed from respiratory and cardiac recordings acquired during scanning. These regressors can account for signal changes related to breathing, cardiac pulsation, and variations in heart rate or respiratory volume. The dataset does not contain such measurements. Therefore, physiological regressors based on external recordings are not included.

The absence of explicit motion or physiological regressors does not imply that the corresponding effects are absent from the data. Some of this variability may remain in the residuals or may later be captured by data-derived nuisance regressors.

Drift regressors

Drift regressors are included to model low-frequency changes in the fMRI signal. Such changes may appear, because of scanner instabilities, slow participant movement, thermal effects, or other slow-varying processes.

Low-frequency drift can be represented using polynomial functions, splines, or Discrete Cosine Transform (DCT) basis functions. In this thesis, a DCT basis is considered. DCT regressors provide a controlled representation of low-frequency variation without over-fitting the drift regressors to the actual desired BOLD signal.

Let n denote the number of time points in a run. A DCT-II basis function [15] can be written as

$$d_k(i) = \cos\left(\frac{\pi k(2i+1)}{2n}\right), \quad i = 0, \dots, n-1, \quad k = 1, \dots, K. \quad (3.74)$$

The constant component corresponding to $k = 0$ is represented separately by an intercept column, and hence will not be included as part of the DCT drift regressors.

The number of DCT regressors should depend on the duration of the run and the selected high-pass cutoff [16]. If T denotes the run duration and τ_c denotes the cutoff period, an approximate rule is

$$K = \left\lfloor \frac{2T}{\tau_c} \right\rfloor. \quad (3.75)$$

Only the low-frequency basis functions below the selected cutoff are included. The cutoff must be chosen carefully because including too many DCT regressors may remove slow task-related variation together with signal drift. For the analyzed dataset, each run lasts approximately 300 seconds, but the number of DCT regressors cannot be determined from the run duration alone. It also requires an explicit choice of the cutoff period τ_c , which in this thesis is the default value provided by SPM [16] $\tau_c = 128s$.

Including the DCT basis in the design matrix is equivalent to modeling the selected low-frequency trends jointly with the task-related regressors. Their fitted contribution can then be treated as part of the nuisance signal and removed according to Equation (3.65).

3.4 PCA-derived nuisance regressors

- This method is inspired by the GLMdenoise method, described in paper [17] - This is an extension of the GLM-based denoising - It utilizes the structure of the experiment, where there are multiple runs, since it utilizes cross-validation (CV)

- The main structure of the algorithm presented in [17] is as follows: 1. After the initial GLM fit for each run, except one, we use this GLM model to predict the left-out run 2. Next comes the selection of noise voxels for the noise pool: - CV R2 value from step 1. must be less, than 0% (preventing voxels related to the experiment from entering the noise pool) - mean signal intensity must be above minimum threshold (preventing voxels outside the brain from entering the noise pool) 3. Calculating noise regressors using PCA on time-series of voxels in the pool

3.5 Independent Component Analysis

3.6 LASSO method

Chapter 4

Application of selected statistical models to the fMRI data

Chapter 5

Summary

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